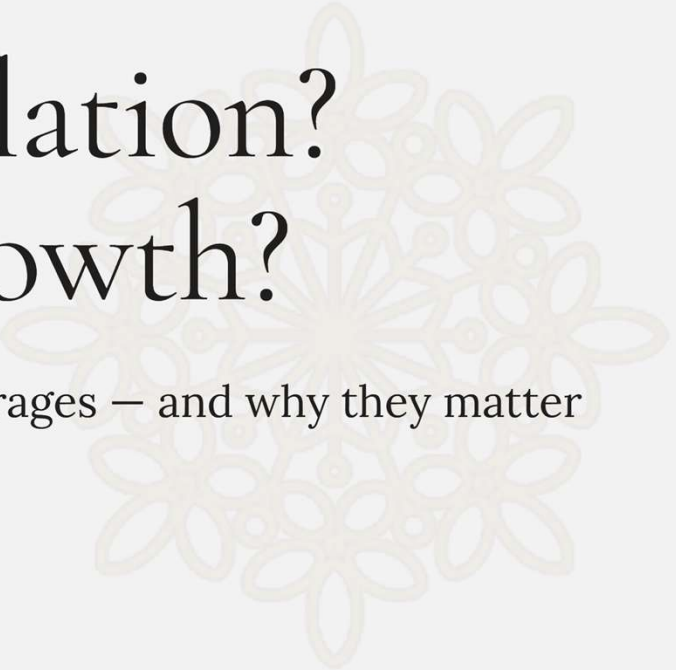
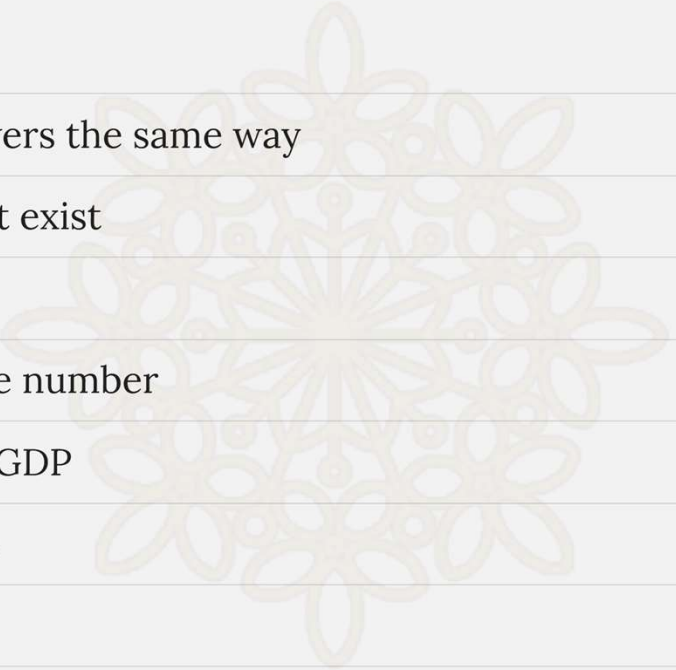


Whose Inflation? Whose Growth?



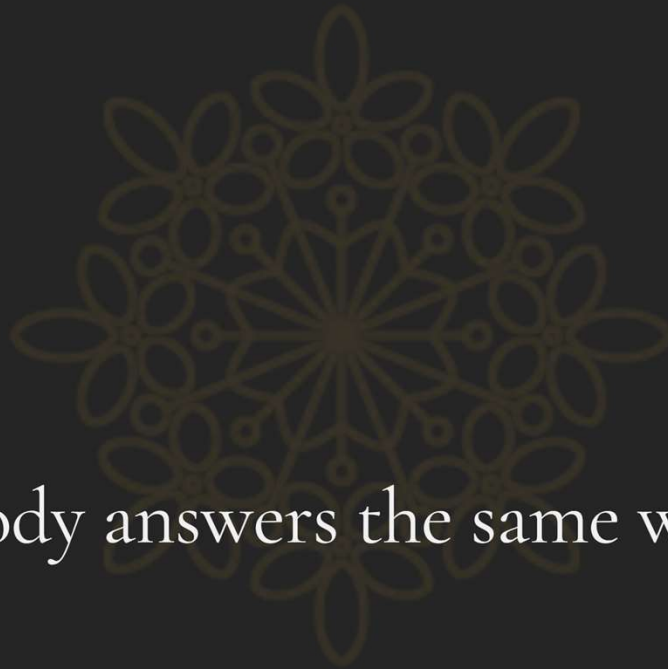
The limits of economic averages — and why they matter
to investors

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O I

The question nobody answers the same way



“What was your inflation rate last year?”

- Did your rent rise? Did your salary rise?
- Did your school or college fees rise?
- Did your domestic help's wages rise?
- Did your Uber or airfare bills rise?
- Did the price of vegetables rise?
- Did your mutual-fund portfolio rise? Your health-insurance premium?
- Did you buy a new laptop or smartphone — or a home?

Official $\neq$ personal $\neq$ perceived inflation

What CPI answers

“How much has the cost of purchasing a statistically representative basket of consumer goods and services changed?”

What CPI does not answer

“How much more expensive has life become for me?”
None of the three numbers — official, personal, perceived — is necessarily wrong.

02

The household that doesn't exist



The representative household

$$\text{CPI} = \sum_{i=1}^n w_i \cdot \Delta p_i$$

w_i is the expenditure weight and Δp_i the price movement of component i . The important question: who is the household represented by those weights? Nobody in particular — it is a statistical construct.

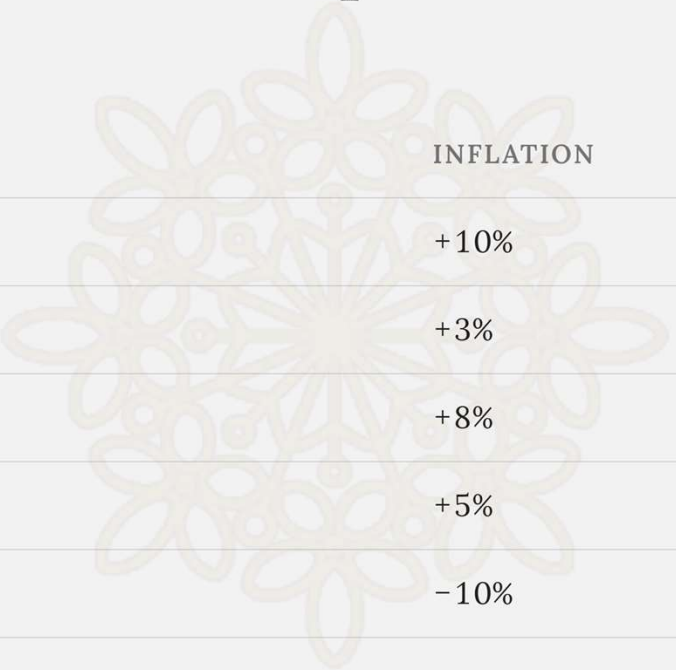
Two households, two baskets



COMPONENT	RURAL WEIGHT	URBAN WEIGHT
Food & beverages	54.18%	36.29%
Housing	no rural weight	21.67%

Source: Ministry of Statistics — 2012 CPI series weight structure.

Same price shock, different experience



COMPONENT	INFLATION
Food	+10%
Rent	+3%
Education	+8%
Transport	+5%
Electronics	-10%

Source: Aggregate CPI of 5% is a particular weighted average — not a description of any one household.

02 · THE HOUSEHOLD THAT DOESN'T EXIST

“If food inflation is 10% and your household spends 15% of its income on food, while another spends 50%, should both households be told they experienced the same inflation?”

O2a

Caselet — India, 2010–13

Food-price pressure fell hardest on households where food is half the budget: the same shock, a very different dominance of the personal basket.

Inflation is a liability-specific phenomenon

High exposure

- A mortgage
- Two children in private school
- Expensive healthcare
- Substantial international travel

Low exposure

- A retired person
- A tenant
- A farmer
- A young professional living with parents

A CPI is a consumption portfolio

INVESTMENT

Portfolio weights

Asset returns

Portfolio return

Benchmark

Tracking error

CONSUMPTION

Consumption weights

Price changes

Personal inflation

CPI

Personal inflation minus CPI

o2b

Caselet — India, 2013–14

Onions: frequently purchased, politically salient, prone to violent swings. A 50% rise here can feel more inflationary than a larger move in something bought once every five years.

03

When the basket itself lies



The substitution problem

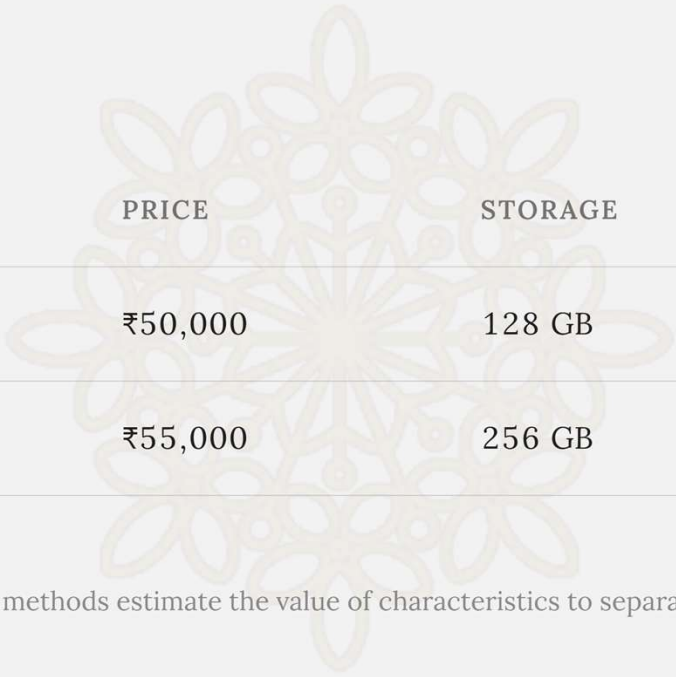
- Apples rise 30%, bananas rise 5% — consumers buy more bananas.
- A fixed basket assumes the consumer keeps buying exactly as before.
- Steak +40%, chicken +15%, dal +5% — a household substitutes downward.
- Should an index price yesterday's lifestyle, or today's standard of living? There is no trivial answer.

Shrinkflation: when the price doesn't rise

- A ₹100 packet becomes 20% smaller while remaining ₹100 — has inflation occurred? Obviously yes, economically.
- The measurement problem: what exactly is the "product" whose price is being tracked?
- Price indices attempt to compare constant-quality, constant-quantity products.

Source: Bureau of Labor Statistics — products disappear, are replaced, and change characteristics.

Hedonic adjustment



	PRICE	STORAGE	CAMERA
Phone A	₹50,000	128 GB	ordinary
Phone B, a year later	₹55,000	256 GB	much better

Source: Bureau of Labor Statistics — hedonic methods estimate the value of characteristics to separate quality gain from pure price change.

03 · WHEN THE BASKET ITSELF LIES

“Who decides how much better the product is?”

Processor speed, camera quality, memory, screen quality, safety, fuel efficiency — a statistical agency has to model the value of each. The number is not simply observed. Some of it is modelled.

The elephant in the room: housing

- If a mortgage payment doesn't change, has housing inflation been zero? Not necessarily.
- The real question: what is the price of the housing *service* consumed?
- The US CPI solves this with Owners' Equivalent Rent — what would this homeowner have paid to rent the equivalent house?

Source: Bureau of Labor Statistics — OER measures the rental value of the shelter service, excluding the investment component.

Asset prices $\neq$ consumption prices

MEASURE

CHANGE

House price

+20%

Mortgage payment

unchanged

Market rent

+5%

Source: CPI's housing concept lands near +5% — it measures consumption of housing services, not the investment return on the house.

03a

Caselet — US, 2021–23

House prices exploded across many US markets, yet CPI does not simply pour the price gain into consumer inflation — a house is simultaneously an asset and a source of consumption services.

04

Why it feels worse than the number



The psychology of the basket

Frequency

Vegetables, petrol, milk, restaurant bills are remembered far more readily than a washing machine or an insurance premium.

Salience

A 20% rise in onions is psychologically louder than a 10% decline in television prices.

Loss aversion

A price increase is noticed far more strongly than an equivalent decrease elsewhere.

The result

Perceived inflation runs persistently above the official number — for reasons that are entirely rational.

04 · WHY IT FEELS WORSE THAN THE NUMBER

9.7%

Median perceived current inflation among urban households surveyed by the RBI in March 2022 — considerably above the official CPI that month; one-year-ahead expectations stood at 10.8%.

Source: Reserve Bank of India — household inflation expectations survey.

Both numbers matter — differently

The CPI affects

- Monetary policy
- Real interest rates
- Indexation and contracts
- Policy decisions

Perceptions affect

- Wage demands
- Consumption and savings
- Expectations
- Pricing behaviour

04a

Caselet — US, 2021

Used cars and trucks up 37.3% year-on-year at end-2021, against shelter up 4.1%. If you don't own a car, should you care? If you urgently need one, is the CPI understating your inflation?

US, 2021–22 — the grand case study

COMPONENT

CHANGE (DEC 2021, YOY)

Headline CPI

+7.0%

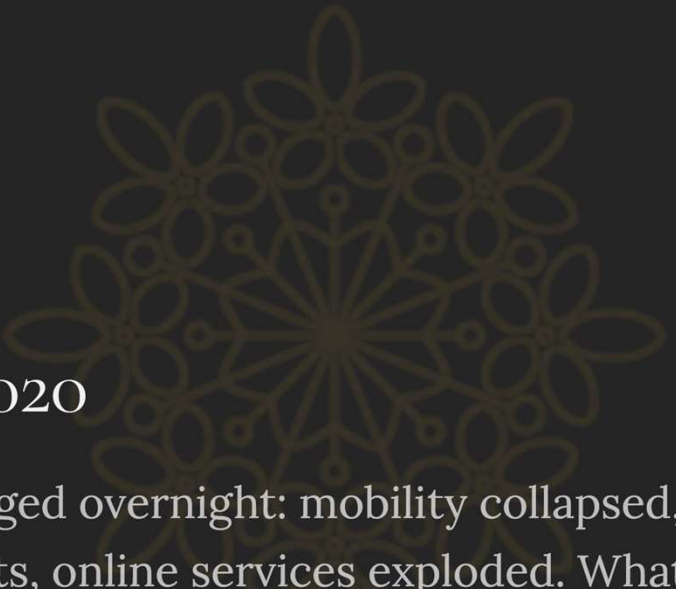
Used cars & trucks

+37.3%

Shelter

+4.1%

Source: Bureau of Labor Statistics, December 2021 CPI release.



o4b

Caselet — India, 2020

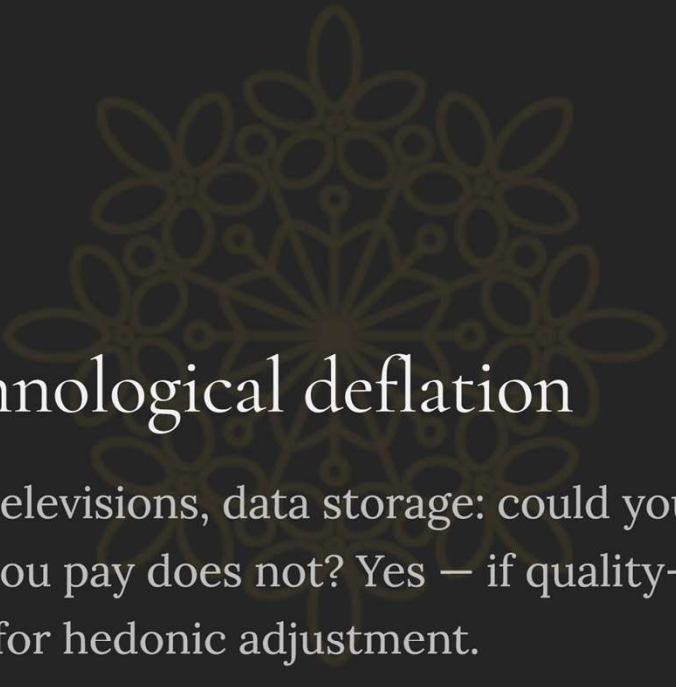
Consumption baskets changed overnight: mobility collapsed, restaurants vanished from many budgets, online services exploded. What does an airline-ticket price tell us if nobody is buying tickets?

O4C

Caselet — India, 2020–22

Rural and urban CPI inflation behaved differently through the pandemic and the inflation episode that followed — why might the same macro shock produce different rates for each?

Source: Reserve Bank of India — rural/urban CPI series.



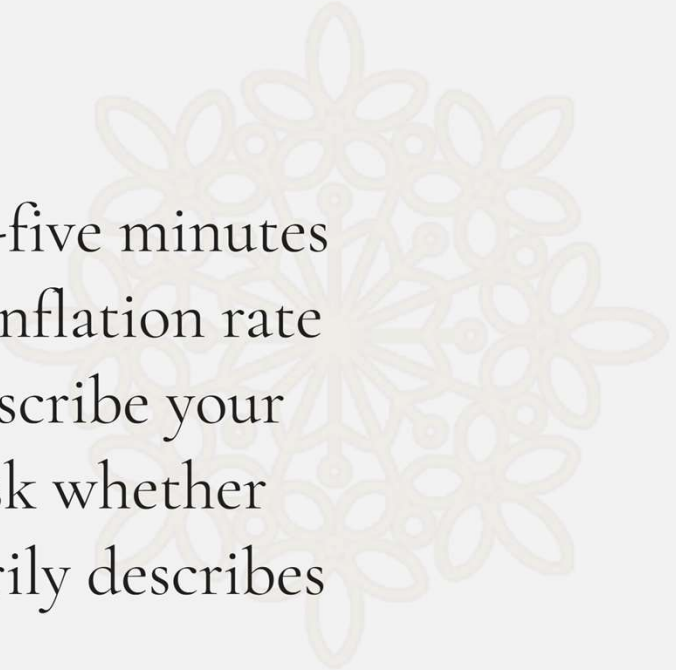
o4d

Caselet — US, technological deflation

Computers, smartphones, televisions, data storage: could your cost of living fall even if the price you pay does not? Yes — if quality-adjusted prices fall. That is the case for hedonic adjustment.

THE PIVOT

“We have spent thirty-five minutes discovering that the inflation rate doesn't necessarily describe your inflation. Now let's ask whether GDP growth necessarily describes your growth.”



05

Now do the same thing to GDP



GDP is also an average

Real GDP per capita growth $\approx$ GDP growth – inflation – population growth

If India's GDP grows 7% while population grows 1%, did the average Indian become 7% richer? No — and that is before considering distribution.



“8% growth” can mean 2% richer

MEASURE

CHANGE

Nominal GDP growth

+8%

Population growth

+1%

Inflation

+5%

Real GDP-per-person growth

≈ +2%

Source: And even 2% does not tell us how the gains were distributed.

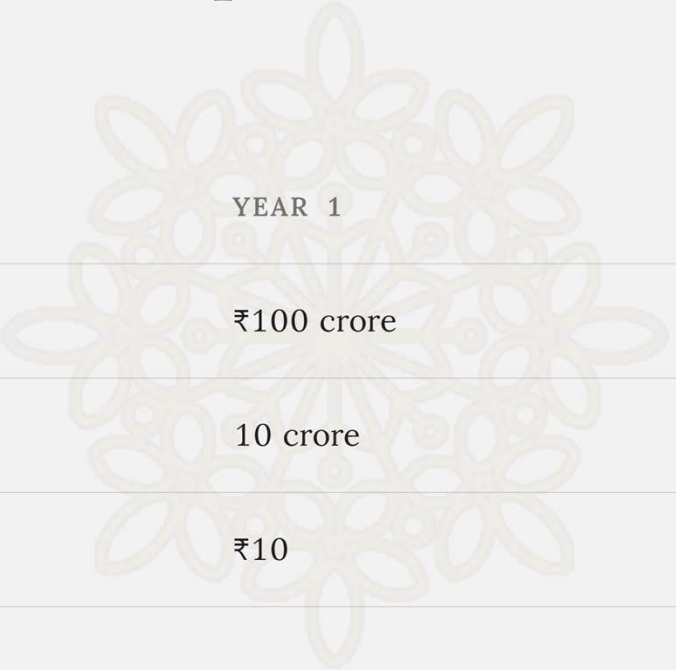
Latticework for Investment Management · Lecture 01

05a

Caselet — US, 2020–21

GDP fell roughly 3% in the pandemic — but is GDP per capita down exactly 3%? Why do we quote GDP growth far more often than GDP-per-capita growth? A question about what makes better headlines.

From per capita GDP to “per shareholder”



	YEAR 1	YEAR 2
Profit	₹100 crore	₹120 crore
Shares outstanding	10 crore	15 crore
EPS	₹10	₹8

Source: Profit grew 20% — EPS fell 20%. Would you buy a stock for total profit growth without asking what happened to the share count?

“GDP dilution” — one hierarchy, two worlds

National level

- GDP
- GDP per capita
- Real GDP per capita
- Median household income
- Distribution of gains

Corporate level

- Profit
- Profit per share (EPS)
- Real / inflation-adjusted EPS
- Median shareholder outcome
- Distribution across shareholders

05 · NOW DO THE SAME THING TO GDP

“Why should an investor care about corporate profit growth but not population growth?”

Denominator discipline is a universal financial principle — it does not stop at the balance sheet.

05b

Caselet — India, 2016

Demonetisation disrupted cash-intensive activity. If a transaction disappears from observable cash channels, has the economic activity disappeared — or merely become harder to measure?

05C

Caselet — India, 2017

GST created a richer digital transaction trail. Can an improvement in measurement make an economy appear to change even when we have simply learned to see it better?



o5d

Caselet — India, 2018–20

The IL&FS and NBFC liquidity stress: if total credit is growing, can a particular class of borrowers simultaneously face a credit crunch?
Obviously yes. Aggregate $\neq$ representative.

06

Who actually got the gains



The distribution problem

- An economy of 100 people. GDP increases 10%.
- The top 10 capture almost all of the increase. The bottom 90 see none.
- Mathematically, the average person is 10% better off. Economically, the statement is deeply incomplete.

06 · WHO ACTUALLY GOT THE GAINS

“If GDP per capita rises 5% but the median household's real disposable income falls 2%, is the economy doing well?”

There is no purely statistical answer — it depends on the question you are asking: aggregate, per capita, median, distribution, real, or nominal.

06a

Caselet — India, 2026

A new national-accounts series (FY2022-23 base) and a new CPI (2024 base, 299 items) both landed this year, alongside 7.8% reported quarterly real GDP growth. If measurement changes, did the economy change — or did our view of it change?

India's 2026 methodology reset

- New CPI: 2024 base year, 299 weighted items.
- New national accounts: FY2022-23 base year, more granular deflators.
- Latest reported quarterly real GDP growth of 7.8% has already sparked debate over methodology and the GDP deflator.
- If the measuring instrument improves, should we be surprised that history gets rewritten? No — that is precisely what revisions are for.

Source: Ministry of Statistics; Reuters — 2026 CPI and national-accounts base revisions.

07

The investor's checklist



Five questions behind every macro number

- **Whose basket?** Who is being represented?
- **What weights?** What does that household actually consume?
- **What quality?** Is the product actually comparable?
- **What denominator?** Per person, per household, per worker, per share?
- **What distribution?** Who actually experienced the change?

The checklist, three ways



QUESTION	CPI	GDP	COMPANY
What is measured?	Cost of consumption	Economic production	Business activity
What are the weights?	Household expenditure	Sectoral composition	Product mix
What is the denominator?	Basket	Population	Shares outstanding
Nominal or real?	Price change	Nominal vs real GDP	Revenue / profit
What about quality?	Hedonic adjustment	Deflators	Product mix
Who benefits?	Household heterogeneity	Distribution	Shareholders
Reality or measurement?	Methodology revisions	Base-year revisions	Accounting reclassification

One principle, four portfolios

CPI

A consumption portfolio.

GDP

A production portfolio.

Your returns

An asset portfolio.

Corporate EPS

A denominator-adjusted profit measure.

“There is no such thing as the inflation rate or the growth rate. There is only a particular measurement of a particular phenomenon, constructed for a particular purpose.”

— the sophisticated investor doesn't ask merely what the number is. He asks what exactly it is measuring — and what it is leaving out.

Discussion

“If two investors can own different portfolios and experience different returns, why should we be surprised that two households can own different consumption portfolios and experience different inflation?”